Networking Academy

Lab - Ohms Law

Answer the following questions based on electricity and Ohm's Law. Show all steps when solving problems.

a. What are the four basic units of electricity? Provide the variable name and symbol, and unit name and symbol.

The four basic units of electricity are Voltage (V), measured in Volts (V), Current (I), measured in Amperes (A), Resistance (R), measured in Ohms (Ω), and Power (P), measured in Watts (W).

b. Write the equation for Ohm's Law.

The equation for Ohm's Law is V = I × R, where voltage is equal to current multiplied by resistance.

c. Re-arrange the Ohm's Law equation to solve the following:

I = *V ÷ R*

R = *V ÷ I*

d. Power is equal to voltage multiplied by current. Add the missing information in each of the following power equations.

P = V *V × I*

P = R *R × I²*

P = V^2 *V² ÷ R*

e. The yellow wire connected to a power supply carries 12V. If the power supply provides 60W of power to the yellow wire, how much current is passing through the yellow wire?

The yellow wire has a voltage of 12 V and a power of 60 W. Using the formula I = P ÷ V, the current is 60 ÷ 12 = 5 A. Therefore, 5 amperes of current are passing through the wire.

f. There are 3.3V passing through an orange power supply cable, and there are 0.25 ohms of resistance in the orange wire. How much power is supplied to the orange wire by the power supply?

The orange wire has a voltage of 3.3 V and a resistance of 0.25 Ω. Using the formula P = V² ÷ R, the power is (3.3²) ÷ 0.25 = 10.89 ÷ 0.25 = 43.56 W. Therefore, the power supplied is 43.56 watts.

g. A wire from the power supply is carrying 120W of power and 24A of current. How much power is supplied to the wire by the power supply?

The wire is carrying 120 W of power and 24 A of current. Using the formula V = P ÷ I, the voltage is 120 ÷ 24 = 5 V. Therefore, the power supply provides 5 volts to the wire.

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